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Multi-Sensor fusion and semantic map-based particle filtering for robust indoor localization

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ABSTRACT

To enhance positioning accuracy and ensure long-term stability in large-scale indoor environments, this paper presents a robust particle filtering method that integrates IMU, magnetic field, Bluetooth, and semantic map data, implemented through a handheld portable device. In the particle prediction phase, based on peak candidates extracted in the time domain, the unintentional interference movement of the hand is removed through short-time Fourier transform in the frequency domain to reduce the noise of pedestrian dead reckoning. During the observation phase, we introduce a magnetic interference coefficient to model and quantify the degree of electromagnetic interference in different environments, thereby improving the accuracy of indoor magnetic headings and enhancing the alignment of particle swarms with real-world directional constraints. Furthermore, based on passable semantic information, the indoor map is segmented into distinct probability regions, allowing the particle filter to adaptively adjust the weight of the particle swarm when moving between regions, thus improving the robustness of the overall fusion system by preventing particles from entering infeasible areas. Experiments and validations were carried out in a large shopping mall and parking lot with multiple devices. Results demonstrate that the proposed method improves the average positioning accuracy by approximately 25% compared to the Bluetooth-only positioning system, with a 15% increase in cumulative distribution probability within 3 m.

1. Introduction

In large-scale indoor environments where satellite signals are unavailable, reliable personnel positioning systems are essential for safety management and navigation. Traditionally, these systems required individuals to wear sensor suites [12] that detected movement and orientation while communicating with remote management or navigation systems, which increased data transmission complexity and reduced real-time responsiveness. However, advancements in microelectromechanical systems (MEMS) and the development of high-performance, low-power processing chips have enabled portable handheld devices, such as smartphones [3] and personal digital assistants (PDA) [4], to handle motion sensing and position estimation independently. As a result, research into integrated indoor positioning methods using handheld smart devices offers substantial practical value, promising enhanced usability and system efficiency.

In recent years, many researchers have explored indoor positioning technologies for GPS-denied environments. Studies [567] show that positioning methods relying on a single sensor often struggle to achieve the accuracy and long-term stability required in large, complex spaces. In contrast, multi-sensor fusion techniques integrate data from various sensors, leveraging their complementary strengths to enhance both accuracy and robustness. Fusion methods can be divided into two main categories: filtering-based and learning-based approaches. Filtering-based methods, such as Extended Kalman Filtering (EKF) [8], Unscented Kalman Filtering (UKF) [9], and Particle Filtering (PF) [10], use kinematic models and noise estimation to fuse data, offering high efficiency in real-time applications. Learning-based methods, on the other hand, employ neural networks for feature extraction and data fusion [111213], which use multi-source sensor data to map real-world locations, optimizing network parameters through backpropagation. However, learning-based methods often struggle with scene-transferability

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and lack the real-time performance needed to support multi-user operations in large, complex environments.

In filtering-based methods, Extended Kalman Filter (EKF) fusion techniques [141516] typically use a first-order approximation of the kinematic model of the positioning object. While this approach increases computational efficiency, its accuracy is often limited in scenarios involving complex motion or high noise. Unscented Kalman Filter (UKF) methods [171819] improve data propagation accuracy in nonlinear models by using multiple sample points based on a Gaussian distribution, but they still struggle with non-Gaussian noise. With advancements in chip processing capabilities, Monte Carlo sampling-based Particle Filtering (PF) techniques have become increasingly popular [20212223]. For example, the method in [21] integrates an improved PF algorithm with geomagnetic map matching and pedestrian dead reckoning (PDR), introducing boundary constraints on particle numbers. In reference [22], a particle swarm optimization algorithm converts Radio Signal Strength Indicator (RSSI) values into distances, followed by path loss model calculations for position estimation. Similarly, reference [23] combines particle swarm optimization with a low-power Bluetooth signal propagation model, improving position tracking accuracy in indoor environments. These studies highlight that particle filtering is highly effective at modeling nonlinear relationships and is particularly well-suited for data fusion in environments with non-Gaussian noise. This capability enhances both the precision and stability of positioning in complex spatial environments.

Modern portable smart devices are typically equipped with MEMS inertial measurement unit (IMU), magnetic sensors, Bluetooth, and Wi-Fi. However, sensors like electromyography (EMG), ultra-wideband (UWB), and RFID, which lack fully standardized interfaces and protocols, are less commonly integrated. Bluetooth Low Energy (BLE) triangulation, which uses signal strength to calculate positions, is a well-established technique [242526]. BLE beacons offer advantages such as ease of deployment and controlled placement. In contrast, inertial and magnetic sensors, which can detect human motion and orientation with high sensitivity, operate with minimal power consumption. Given these features, it is crucial to explore IMU/Magnetic/BLE data fusion methods for handheld devices, particularly in large and complex indoor environments. Such research has great potential to significantly improve positioning accuracy and operational efficiency.

Among of all, IMU-based positioning technologies include Pedestrian Inertial Navigation Systems (PINS) [27], Pedestrian Dead Reckoning (PDR) [8], and data-driven learning algorithms [28]. PINS typically use foot-mounted sensors to correct inertial drift through zero-velocity updates (ZUPT). However, accurately applying this technique to handheld devices is challenging, as detecting zero velocity is difficult. Data-driven methods, which train neural networks to predict pedestrian speed, can be computationally intensive and time-consuming on portable devices. In contrast, PDR offers a more practical solution by estimating movement displacement through step detection and step length estimation. It is versatile, supporting various mounting positions, including handheld, waist-mounted, and head-mounted configurations, and exhibits strong generalization across different scenarios and multi-user environments. This adaptability makes PDR particularly suitable for portable devices in dynamic indoor settings.

Magnetic field-based indoor positioning methods typically fall into two categories: magnetic heading-assisted localization [2930] and magnetic sequence matching [3132]. The former relies on established principles but is highly susceptible to electromagnetic interference in environments, leading to significant errors due to variations in electromagnetic conditions across different settings. The latter involves creating an offline geomagnetic map that records magnetic field distortions caused by reinforced concrete structures in indoor spaces. During the online phase, positioning is achieved by matching sequences of continuous magnetic field samples. However, in large indoor areas, the distinctiveness of these magnetic sequences weakens, reducing the effectiveness of this approach.

In autonomous driving and robotics, significant research has focused on creating and interpreting semantic maps. For instance, reference [33] uses a semantic-aware interactive prediction method to capture spatial information and objects, enabling more accurate trajectory prediction for autonomous vehicles. In reference [34], semantic information is integrated into Normal Distribution Transform (NDT)-based mapping, significantly improving the real-time speed and robustness of robot mapping. Similarly, reference [35] develops a reliable long-term 3D visual semantic map by combining a 2D semantic plane with visual tag observations. These studies demonstrate that semantic information provides strong discriminative power and long-term stability, essential for reliable positioning in large-scale environments.

Building on the research background and existing approaches, this paper presents a particle filtering method that integrates multi-sensor fusion and semantic map constraints to improve positioning accuracy and robustness in large, complex indoor environments. In the particle prediction phase, an enhanced step detection technique combines time-domain peak detection with frequency-domain filtering to eliminate unintentional hand movements. During the observation phase, an adaptive magnetic interference coefficient is introduced to quantify electromagnetic noise in various indoor environments, enhancing the accuracy of magnetic heading constraints. Additionally, the indoor environment is modeled using passable semantic information, where different regions are assigned probabilistic constraints to guide particle movement, further increasing the system robustness.

The main contributions of this study are:

1) a particle filtering framework that integrates IMU, magnetic sensors, BLE, and semantic maps;

2) improved step detection method that combines time and frequency domain techniques to minimize unintentional movements;

3) enhanced magnetic heading method using interference coefficient modeling for greater accuracy;

4) semantic maps modeling and constraint based on passable probabilities.

2. System scheme

The system scheme of the proposed method is shown in Fig. 1. It employs a particle filtering framework to integrate data from IMU, magnetic sensors, Bluetooth, and semantic maps, which is a recursive filter based on the Monte Carlo method [2036]. The position and heading are estimated by computing a weighted average over a set of particles, where each particle represents a possible pedestrian state under the distribution of observation noise. The state of each particle is defined by three variables $\begin{bmatrix} p_x & p_y & \theta \end{bmatrix}^T$, along with a corresponding weight w.

Initially, the positions and headings of the particles are initialized using Bluetooth signals and magnetic heading information. The subsequent particle prediction step is carried out based on the Pedestrian Dead Reckoning (PDR) algorithm, where unintentional hand movement interferences are detected and filtered out. Observations, including data

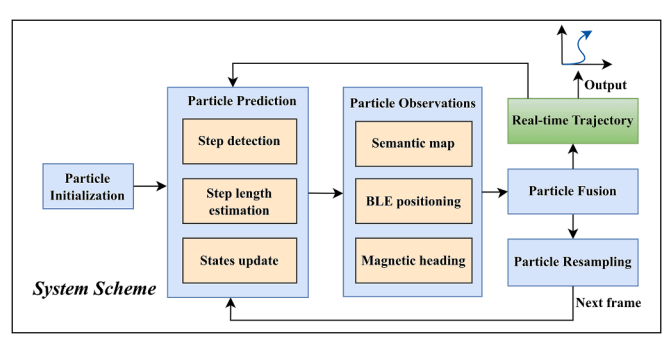

Fig. 1. System scheme.

from semantic maps, Bluetooth, and magnetic field, are then used to update and correct the particle weights. Finally, the current position and heading of the handheld device are estimated by fusing the states and weights of all particles. Additionally, before each new frame of PDR-based particle prediction, the short-term walking step length is dynamically estimated using a smoothed real-time trajectory derived from the previous data.

The coordinate system of the handheld device is defined as depicted in Fig. 2. The *b*-frame represents the body coordinate frame of the device, which also functions as the measurement coordinate frame for the sensors, with its origin at the center of gravity. In the frame, the *x, y* and *z* axes correspond to the forward, left, and upward directions relative to the device. The *h*-frame denotes the horizontal coordinate frame, whose origin coincides with the *b*-frame. In this frame, the *x, y* and *z* axes align with the forward, left, and upward relative to the current horizontal plane.

2.1. Particle initialization

The initialization of particle positions and headings is performed using Bluetooth localization and magnetic heading, each modeled respectively by 2D and 1D Gaussian distributions, respectively, as shown in Fig. 3. The mean of the initial particle position distribution is obtained from the first-frame position provided by the Bluetooth-based positioning system, while the initial variance is determined from the statistical results of the Bluetooth positioning accuracy. Similarly, the mean and variance of the initial heading distribution are derived from the first-frame magnetic heading and the statistical characteristics of the magnetic system. Particles are used to approximate the statistical distribution of the true position of the target. The greater the number of particles, the more accurately the statistical characteristics approximate the true position. However, each iteration of motion prediction and observation correction in particle filtering requires individual computation for each particle. Therefore, a larger number of particles enhances system performance and robustness but requires higher computational complexity. In practice, the number N of particles can be adjusted based on processor capabilities. To enhance the overall robustness of the system, initial particle weights are set to equal values, which will slightly increase the initial convergence time.

2.2. Particle prediction

Pedestrian Dead Reckoning (PDR) serves as the primary algorithm for particle prediction. The core procedures of the PDR algorithm can be divided into three key components: step detection, which identifies walking steps based on IMU data; step length estimation, which calculates the distance covered per step; and the continuous updating of heading and position to track the pedestrian's movement.

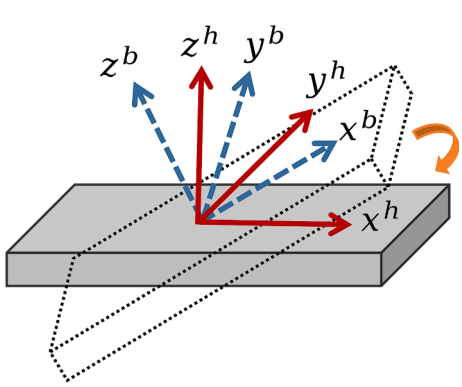

Fig. 2. The coordinate system of the handheld device.

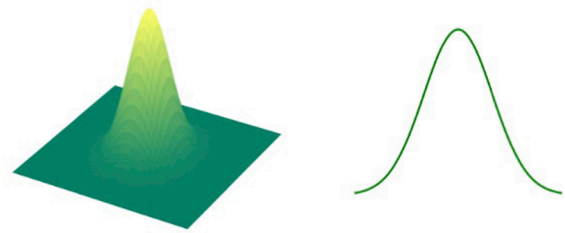

Fig. 3. The Initial distribution of position(left) and heading(right).

2.2.1. Step detection

Step detection is performed using the signal from the z-axis accelerometer in the horizontal coordinate system (*h*-frame), which combines both time-domain and frequency-domain characteristics as depicted in Fig. 4. Firstly, the accelerometer data from the body coordinate system (*b*-frame) is transformed into the *h*-frame to obtain tilt-compensated acceleration. Next, an adaptive window peak detection method is applied to the z-axis acceleration in the *h*-frame to identify potential steps, resulting in undetermined step intervals [3738]. Subsequently, the Short-Time Fourier Transform (STFT) analyzes the frequency spectrum of these intervals, filtering out results that do not align with the typical walking frequency (1–3 Hz). This combined approach refines step detection, producing accurate step timestamps and well-defined step intervals.

Equation (1) expresses the coordinate transformation of acceleration data, where $\begin{bmatrix} a_x^h & a_y^h & a_y^h \end{bmatrix}^T$ and $\begin{bmatrix} a_x^b & a_y^b & a_y^b \end{bmatrix}^T$ represent the three-axis components of accelerometer in *h*-frame and *b*-frame, respectively. $\begin{bmatrix} a_{xbias}^b & a_{ybias}^b & a_{ybias}^b \end{bmatrix}^T$ denotes the three-axis bias of accelerometer, which can be calibrated offline through a turntable. C_b^h is the rotation matrix from *b*-frame to *h*-frame, calculated as follows:

$$a^h = \begin{bmatrix} a_x^h \\ a_y^h \\ a_z^h \end{bmatrix} = C_b^h \bullet \left(\begin{bmatrix} a_x^b \\ a_y^b \\ a_z^b \end{bmatrix} - \begin{bmatrix} a_{xbias}^b \\ a_{ybias}^b \\ a_{zbias}^b \end{bmatrix} \right) \tag{1}$$

$$C_b^h = R(\alpha)^T R(\beta)^T$$

$$= \begin{bmatrix} cos(\alpha) & 0 & -sin(\beta) \\ sin(\alpha) \bullet sin(\beta) & cos(\alpha) & sin(\alpha) \bullet cos(\beta) \\ cos(\alpha) \bullet sin(\beta) & -sin(\alpha) & cos(\alpha) \bullet cos(\beta) \end{bmatrix} \tag{2}$$

$$R(\alpha) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\alpha) & -\sin(\alpha) \\ 0 & \sin(\alpha) & \cos(\alpha) \end{bmatrix}, R(\beta) = \begin{bmatrix} cos(\beta) & 0 & sin(\beta) \\ 0 & 1 & 0 \\ -sin(\beta) & 0 & cos(\beta) \end{bmatrix} \tag{3}$$

where $R(\alpha)$ and $R(\beta)$ are the direction cosine matrices corresponding to rotations about the x and y axes. α denotes the roll angle (rotation around the x-axis), and β represents the pitch angle (rotation around the y-axis).

2.2.2. Step length estimation

Existing step length mathematical models [163038] often require offline training of parameters and make predictions and estimations based on IMU data during online operation. Due to individual differences, this method may result in lower accuracy for multi-user scenarios and complex handheld conditions. During walking, the step length tends

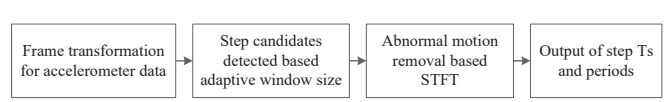

Fig. 4. Multimodal step detection.

to remain relatively stable over short intervals, enabling the prediction of the current step length of the particles based on the historical movement trajectory. To introduce diversity and account for individual differences, the step length (SL) prediction model is represented as a Gaussian probability distribution, as expressed in Equation (4):

$$\widetilde{sl} \sim N(\mu_{sl}, \sigma_{sl}^2) \tag{4}$$

where the mean and variance of the model are calculated based on the short-term historical trajectory provided by the particle filter and the step detection results.

$$\mu_{sl} = \frac{\Delta s(t_{k-1} - \tau, t_{k-1})}{M}, \sigma_{sl} = \sqrt{\frac{1}{M}\sum(\widehat{sl}_i - \mu_{sl})^2} \tag{5}$$

$$\widehat{sl}_i = \Delta s(t_{i-1}, t_i), i \in (t_{k-1} - \tau, t_{k-1}) \tag{6}$$

In Eqs. (5) and (6), $\Delta s(t_k - \tau, t_k)$ denotes the cumulative trajectory length from the particle filter during the time interval $(t_k - \tau, t_k)$. M represents the number of steps within this short-term interval, and t_k indicates the current k-th step moment. τ refers to the length of the short-term historical interval, generally set to a few seconds. $\widehat{sl}_i$ is the single step length during $(t_{k-1} - \tau, t_{k-1})$ in the short-term historical trajectory.

2.2.3. States update

The states update of each particle includes two components: heading update and position update. By integrating the value of the z-axis gyroscope in h-frame over time, heading update can be achieved:

$$\widetilde{\theta}_k = \widetilde{\theta}_{k-1} + \int_{t_{k-1}}^{t_k} \omega_z^h dt \tag{7}$$

where $\widetilde{\theta}_{k-1}$ represents the heading angle of each particle at the end of the previous step. The z-axis gyroscope component ω_z^h in h-frame can be calculated as shown:

$$\begin{bmatrix} \omega_x^h \\ \omega_y^h \\ \omega_z^h \end{bmatrix} = C_b^h \bullet \left(\begin{bmatrix} \omega_x^b \\ \omega_y^b \\ \omega_z^b \end{bmatrix} - \begin{bmatrix} \omega_{xbias}^b \\ \omega_{ybias}^b \\ \omega_{zbias}^b \end{bmatrix} \right) \tag{8}$$

where C_b^h indicates the rotation matrix from b-frame to h-frame (refers to Equation (2)), and $\begin{bmatrix} \omega_x^b & \omega_y^b & \omega_y^b \end{bmatrix}^T$ represents the three-axis data of gyroscope of handheld device in b-frame. $\begin{bmatrix} \omega_{xbias}^b & \omega_{ybias}^b & \omega_{ybias}^b \end{bmatrix}^T$ denotes the three-axis bias of gyroscope. Gyroscope drift can introduce errors in the heading of the PDR algorithm over time, affecting the position estimation. Initial zero bias of the gyroscope can be effectively compensated using the standing calibration method published in prior work of our team [39]. To address the heading drift after system start-up, observational constraints from Bluetooth, magnetic, and semantic maps are applied to suppress the growth of errors.

Based on the position of each particle at the moment of k-1, along with the predicted k-th step length and heading angle, the k-th position of each particle can be updated iteratively as follows:

$$\begin{bmatrix} \widetilde{p}_x \\ \widetilde{p}_y \end{bmatrix}_k = \begin{bmatrix} 1 & 0 & \widetilde{sl}_k \bullet cos\left(\widetilde{\theta}_k\right) \\ 0 & 1 & \widetilde{sl}_k \bullet sin\left(\widetilde{\theta}_k\right) \end{bmatrix} \begin{bmatrix} \widetilde{p}_x \\ \widetilde{p}_y \\ 1 \end{bmatrix}_{k-1} \tag{9}$$

2.3. Particle observations

After predicting particle motion using the PDR algorithm, additional observational constraints are applied based on semantic maps, Bluetooth signals, and magnetic heading. Although Bluetooth and magnetic signals are widely used for indoor positioning, Bluetooth are prone to interference and multipath effects, while magnetic signals can be distorted by large electromagnetic devices. These limitations pose significant challenges in maintaining stable positioning across diverse indoor environments.

To address these challenges and improve the robustness of indoor positioning estimates, this paper proposes two key enhancements: 1) introducing motion constraints for particles derived from semantic maps, which incorporate passable probabilities of indoor spaces, and 2) optimizing magnetic heading constraints by integrating a magnetic interference coefficient that quantifies local electromagnetic disturbances. Detailed implementation is described below.

2.3.1. Semantic map

Based on the varying passable probabilities of different indoor locations, the environment is divided into three categories: (a) fully passable area, (b) semi-passable area, and (c) impassable area. As shown in Fig. 5, the fully passable areas include open corridors or pathways, indicated by the white regions connecting entrance and exit points. Semi-passable area, outlined in green line, represent spaces such as offices or fire exits, which are accessible under specific conditions. The impassable areas, marked in red line, encompass walls and restricted zones, such as hollow sections or non-navigable spaces. When particles traverse these different regions, they are assigned distinct passable probability models based on their location. These models dynamically adjust the weights of particles, allowing the system to reflect realistic movement constraints within the environment:

$$\begin{cases} w_{update} = w \bullet p_a \in a \\ w_{update} = w \bullet p_b, \in b \\ w_{update} = w \bullet p_c, \in c \end{cases} \tag{10}$$

where a,b,c represent different regions, and p_a, p_b,p_c denote the passable probabilities, with $p_a > p_b > p_c$. w refers to the original weights of each particle, and w_{update} represents the updated weights after being corrected by the passable semantic map.

2.3.2. Bluetooth signals

Bluetooth signals provide positional constraint on particle movement by using the signal attenuation model [40] and the received Receive Signal Strength Indication (RSSI) values. Based on these RSSI, the distance d between the current handheld device and each Bluetooth base station can be estimated. The normalized weights L_{wj} for each base station are calculated with the weight inversely proportional to the distance:

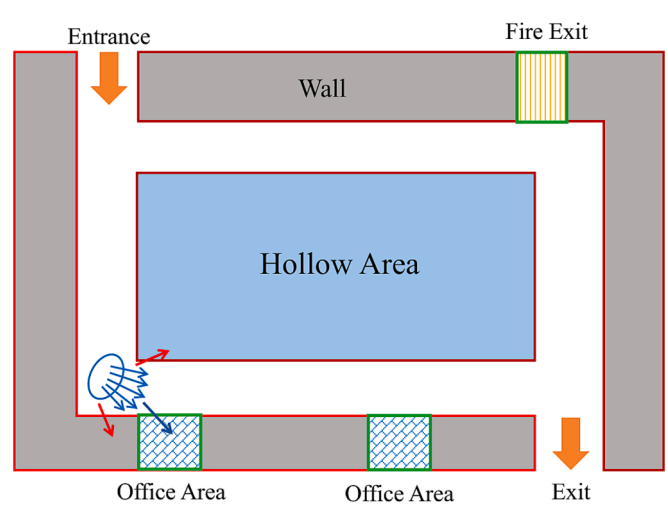

Fig. 5. Semantic map constraints.

$$d = 10^{\left(\frac{\|RSSI-A\|}{10\lambda}\right)} [40] \tag{11}$$

$$L_{wj} = \frac{1}{d_j} \Big/ \sum \frac{1}{d_j} \tag{12}$$

where λ indicates the environmental attenuation factor. A represents the RSSI value of the handheld device at the distance of 1 m from the Bluetooth station, serving as a reference for distance calculation. j is the serial number of base stations.

Subsequently, relying on these weights and the coordinates of the base stations, the Bluetooth positioning result p_{ble} can be obtained:

$$p_{ble} = \sum \left(L_{wj} \bullet Locator_j\right) \tag{13}$$

where $Locator_j$ denotes the coordinates of the j-th base station.

Then, based on computed Bluetooth observation error Δp and the Bluetooth positioning error probability model $P(\Delta p)$, positional constraints are applied to each particle by adjusting their weights:

$$w_{update} = \frac{1}{\sqrt{2\pi} \bullet \sigma_{ble}} \exp\left(-\frac{(\Delta p - e_{ble})^2}{2\sigma_{ble}^2}\right) \tag{14}$$

$$\Delta p = \|p_{ble} - p_i\|_2 \tag{15}$$

$$P(\Delta p) \sim N\left(e_{ble}, \sigma_{ble}^2\right) \tag{16}$$

where w_{update} denotes the new weight of each particle after adding the Bluetooth constraint. Δp represents the Bluetooth positioning error, while p_{ble} and p_i indicate Bluetooth positioning result and the current position of the i-th particle, respectively. $P(\Delta p)$ is the Gaussian model of the Bluetooth positioning error, established via offline evaluation based on the Lidar SLAM (Simultaneous Localization and Mapping) system in the current environment, with e_{ble} and σ_{ble}^2 representing the mean and variance of the model.

2.3.2. Magnetic heading

Similar to the position constraints provided by Bluetooth, magnetic sensor offers heading constraint for particle motion. The noise of magnetic heading data varies across different scenarios due to the influence of various electromagnetic devices [29]. Therefore, a magnetic interference coefficient is introduced to adjust the magnetic constraint model. The adjusted magnetic heading constraint model is as follows:

$$w_{update} = \frac{1}{\sqrt{2\pi} \bullet \varepsilon \bullet \sigma_{mag}} \exp\left(-\frac{(\Delta\theta - e_{mag})^2}{2(\varepsilon \bullet \sigma_{mag})^2}\right) \tag{17}$$

$$\Delta\theta = \|\theta_{mag} - \theta_i\|_2 \tag{18}$$

$$P(\Delta\theta) \sim N\left(e_{mag}, \sigma_{mag}^2\right) \tag{19}$$

where $\Delta\theta$ denotes the magnetic heading error, while θ_{mag} and θ_i represent the current magnetic heading and the heading state of each particle. $P(\Delta\theta)$ is the Gaussian model of the magnetic heading error, constructed through evaluation based on the Lidar SLAM system in the current environment, with e_{mag} and σ_{mag}^2 as the mean and variance, similarly.

The magnetic interference coefficient ε characterizes the intensity of electromagnetic interference in the current environment. A larger value indicates greater instability of the magnetic heading observation. During the offline evaluation phase, ε is determined by the quality of fit of the Gaussian model to the magnetic heading error.

$$\varepsilon = 1 + \sqrt{\frac{1}{n-1} \sum \|(P(\Delta\theta) - f(\Delta\theta)\|^2} \tag{20}$$

where $P(\Delta\theta)$ refers to the probability density of the Gaussian model for magnetic heading errors, and $f(\Delta\theta)$ represents the frequency of magnetic errors in the offline evaluation data. n is the number of evaluated frames of the data.

Additionally, after updating the weights of particles based on this observation, normalization operation is applied to ensure that unitization of the sum of particles weights:

$$w_{norm}(i) = w_{update}(i) \Big/ \sum w_{update}(i) \tag{21}$$

where $w_{update}(i)$ and $w_{norm}(i)$ respectively denote the weight of each particle before and after normalization.

2.4. Particle fusion

Finally, based on the normalized weights assigned to each particle, the fused position and heading are computed through a weighted sum. This process aggregates the information from all particles, ensuring that those with higher likelihoods contribute more significantly to the final estimate:

$$\begin{bmatrix} \widehat{p}_x \\ \widehat{p}_y \\ \widehat{\theta} \end{bmatrix} = \sum_{i=0}^{N} w_{norm}(i) \bullet \begin{bmatrix} \widetilde{p}_{xi} \\ \widetilde{p}_{yi} \\ \widetilde{\theta}_i \end{bmatrix} \tag{22}$$

where $\begin{bmatrix} \widehat{p}_x & \widehat{p}_y & \widehat{\theta} \end{bmatrix}^T$ represents the final fused position and heading at the current moment. $\begin{bmatrix} \widetilde{p}_{xi} & \widetilde{p}_{yi} & \widetilde{\theta}_i \end{bmatrix}^T$ and $w_{norm}(i)$ respectively denote the position, heading angle, and weight of the i-th particle. N is the total number of particles.

After each filtering process, to prevent particle swarm degeneracy, a resampling step is performed. If the number of effective particles is low, the position and heading of particles are resampled with the roulette wheel method according to their weights [4142]. The weights of the resampled particles are set to equal values.

3. Experiments and analysis

3.1. Setup and equipment

To validate the proposed method, experiment data were collected from various handheld smartphone devices in both a parking lot and a shopping mall. These data include readings from low-cost, built-in smartphones sensors such as accelerometers, gyroscopes, magnetometers, and Bluetooth. Simultaneously, a backpack-mounted LiDAR SLAM system was employed to collect raw data, including outputs from a LiDAR (Velodyne VLP-16) and an IMU (ADIS-16470), as depicted in Fig. 6. After processing this data using a SLAM algorithm [43], centimeter-level positioning trajectories and point cloud maps can be obtained. The trajectory derived from Lidar SLAM system serves as the ground truth for evaluating the positioning accuracy of the handheld devices.

3.2. Evaluation indicators

The Cumulative Distribution Function (CDF) and the Average Positioning Error (APE) are employed to evaluate and analyze the proposed method. The CDF of positioning errors can illustrate the proportion of errors that fall below a specified threshold in the entire dataset, as expressed in Equation (23). To facilitate a quantitative comparison of accuracy, three typical positioning error thresholds of 3 m, 5 m, and 10 m are selected for analysis, and the corresponding statistical results are denoted as CDF(3), CDF(5), and CDF(10):

$$CDF(\zeta) = \frac{Sum(Err < \zeta)}{T_n} \tag{23}$$

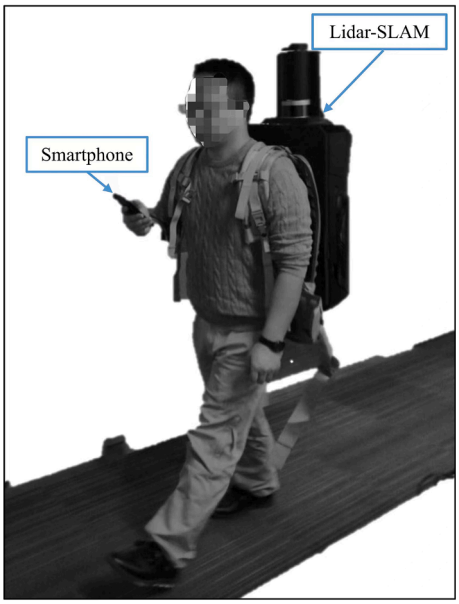

Fig. 6. Experiment equipment.

where $Sum(err < \zeta)$ represents the number of frames in which the positioning error is less than the threshold, and T_n is the total number of positioning frames.

The Average Positioning Error (APE) describes the overall average performance of the test data and is calculated as follows:

$$APE = \frac{1}{T_n}\sum \|Err\| \tag{24}$$

where Err donotes the positioning error of single frame, which can be calculated with:

$$Err = \sqrt{(p_x - gt_x)^2 + \left(p_y - gt_y\right)^2} \tag{25}$$

where $\left(p_x, p_y\right)$ represents the positioning result of proposed method, while $\left(gt_x, gt_y\right)$ indicates the ground truth from the Lidar SLAM system.

3.3. Scenes validation

The validation scenes include a large underground parking lot and a shopping mall as shown in Fig. 7, both of which present significant challenges. The parking lot's difficulties stem from: 1) the substantial contrast between occupied and vacant states; and 2) the potential for Bluetooth signal interference caused by the extensive open spaces. In the shopping mall, the challenges are: 1) high pedestrian density, which leads to increased unintended movements such as stopping to observe or converse; and 2) a complex electromagnetic environment due to the presence of numerous electronic devices. These scenarios effectively represent typical challenges encountered in complex indoor positioning systems.

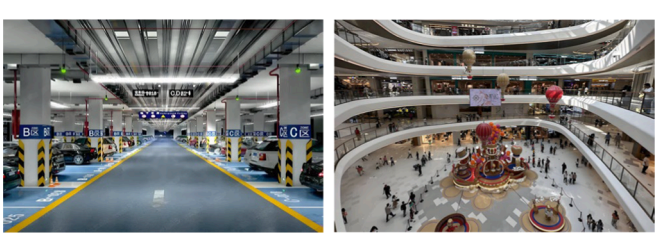

Fig. 7. The scenes of parking lot (left) and shopping mall (right).

In the following test results, MS-PF denotes the multi-sensor fusion and semantic map-based particle filter method described in this paper. BLE represents the Bluetooth-only positioning method, and GT stands for the ground truth provided by the Lidar SLAM system. The collected data from different handheld smartphone device are represented as Sequence-**(Seq-**). Path Length indicates the total length of the walking trajectory.

3.3.1. Parking lot

Table 1 shows the CDF and APE error statistics for the proposed method (MS-PF) and the Bluetooth-only positioning method (BLE) in the parking lot environment. The cumulative path length of all data sequences is 6325.34 m, with each handheld device having a path length exceeding 700 m. The APE for the BLE method is 3.17 m, while the APE for the proposed MS-PF method is 2.12 m, representing a 33.12 % reduction. The CDF(3), CDF(5), and CDF(10) values for BLE are 60.88 %, 83.75 %, and 96.80 %, respectively, whereas the proposed method achieved 81.72 %, 95.44 %, and 99.56 %, indicating improvements of 20.84 %, 11.69 %, and 2.76 %, respectively.

Fig. 8 illustrates the positioning trajectory and the CDF error curve for the handheld device of Seq-06 in the parking lot. The blue dot-and-dash line represents the positioning trajectory of the BLE method, and the red solid line represents the trajectory of the proposed MS-PF method. The black dashed line denotes the reference ground truth trajectory from the Lidar SLAM system. As shown in the results, MS-PF method exhibits reduced positioning noise and smoother trajectories compared to the BLE method. In the CDF curve, the distribution of the MS-PF method is higher than that of the BLE positioning method for various error values.

3.3.2. Shopping mall

Table 2 presents the positioning error statistics for the shopping mall environment, where nine different handheld devices from various manufacturers were tested. The path length for each device varied between approximately 400 to 700 m, resulting in a total cumulative path length of 5620.75 m. The APE for the BLE method is 2.51 m, while the APE for the proposed MS-PF method is 1.88 m, representing a reduction of 25.10 %. The average CDF values for the BLE method are 69.16 % for CDF(3), 93.04 % for CDF(5), and 99.68 % for CDF(10). In the proposed MS-PF method, the average CDF values are 86.36 %, 97.53 %, and 99.95 %, showing improvements of 17.20 %, 4.49 %, and 0.27 %, respectively.

Fig. 9 shows the positioning trajectory and the CDF curve for Seq-08 in the shopping mall. The results indicate that the positioning error of the MS-PF is significantly smaller. And compared to the BLE method, the MS-PF method shows higher proportions in the CDF curve, consistent with the findings from the parking lot environment.

3.4. Ablation studies

To validate the effectiveness of the key modules in the proposed particle filter method, we conducted ablation studies on the multimodal step detection, semantic map constraints, and adaptive magnetic constraint modules. In these experiments, we employed a variable-controlling method to selectively remove or disable critical modules while maintaining the same environmental conditions for performance comparison. This approach enabled us to isolate and evaluate the individual contribution of each module to the overall system performance, providing a clear assessment of their impact on key positioning metrics.

3.4.1. Multimodal step detection

Table 3 shows the positioning results of the multimodal step detection algorithm in the shopping mall. TD represents the time-domain-based step detection method, while TD & FD refers to the method proposed in this paper, which combines time-domain and frequency-domain information. Compared to the TD method, the proposed

Table 1
The positioning results of parking lot.

Phone Type	Path Length /m	CDF(3) /%		CDF(5) /%		CDF(10) /%		APE /m	
		BLE	MS-PF	BLE	MS-PF	BLE	MS-PF	BLE	MS-PF
Seq-01	1018.00	55.04	74.55	81.10	91.05	96.81	98.98	3.44	2.38
Seq-02	1016.08	50.28	75.33	73.35	92.09	89.6	98.68	4.44	2.83
Seq-03	751.64	49.22	72.57	73.83	93.75	94.63	99.65	3.89	2.54
Seq-04	750.43	80.26	92.31	95.38	98.72	99.74	99.74	1.95	1.39
Seq-05	752.66	64.17	85.33	87.67	97.83	99.50	100	2.84	1.93
Seq-06	1018.06	63.97	86.39	88.85	97.18	98.72	99.87	2.76	1.86
Seq-07	1018.47	63.25	85.53	86.04	97.44	98.59	100	2.87	1.90
Total/Mean	6325.34	60.88	**81.72**	83.75	**95.44**	96.80	**99.56**	3.17	**2.12**

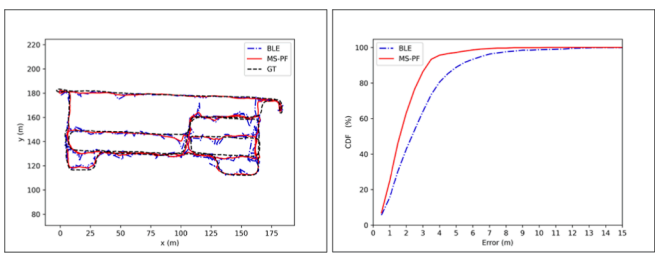

Fig. 8. The trajectory (left) and CDF (right) of Seq-06 in parking lot.

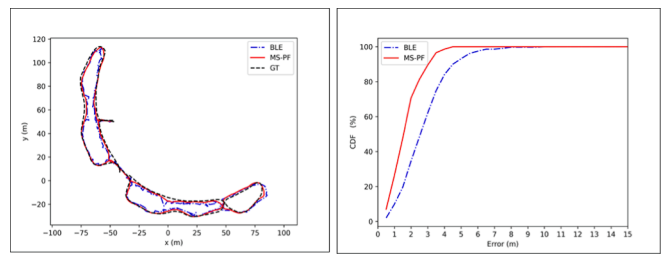

Fig. 9. The trajectory (left) and CDF (right) of Seq-08 in shopping mall.

method achieved a 1.23 % improvement in CDF(3) and reduced the average positioning error (APE) by 0.04 m.

3.4.2. Semantic map constraints

Table 4 presents the positioning results in the shopping mall environment with and without semantic map constraints (No SC and with SC). The APE decreased by 0.37 m, and the cumulative distribution function values CDF(3), CDF(5), CDF(10) improved by 9.40 %, 1.43 %, 0.06 %, respectively. These results clearly demonstrate a significant improvement in positioning accuracy when semantic map constraints are incorporated.

3.4.3. Adaptive magnetic constraint

To assess the efficacy of the adaptive magnetic constraint across various environments, we applied the parameters $[\varepsilon, e_{mag}, \sigma^2_{mag}]$ derived from the mall environment to other scenarios, including the office and parking lot. This approach is denoted as the Fixed Constraint Method (FCM). In contrast, the Adaptive Constraint Method (ACM) utilized magnetic heading constraint parameters that were dynamically evaluated for each specific scenario. The results in Table 5 indicate that, compared to the FCM, the adaptive approach improved the mean of cumulative distribution function values CDF(3), CDF(5), CDF(10) by 1.86 %, 2.02 %, 0.02 %, while decreasing the APE by 0.06 m.

Table 3
The ablation of step detection module.

Mall	CDF(3) /%	CDF(5) /%	CDF(10) /%	APE /m
TD	85.13	97.32	99.85	1.92
TD & FD	**86.36**	**97.53**	**99.95**	**1.88**

Table 4
The ablation of semantic map constraint.

Mall	CDF(3) /%	CDF(5) /%	CDF(10) /%	APE /m
No SC	76.96	96.10	99.89	2.25
With SC	**86.36**	**97.53**	**99.95**	**1.88**

Table 5
The ablation of adaptive magnetic constraint.

Methods	Scenes	CDF(3) /%	CDF(5) /%	CDF(10) /%	APE /m
FCM	Office	83.20	94.10	99.95	1.92
	Mall	86.36	97.53	99.95	1.88
	Parking	80.55	93.20	99.56	2.18
	Mean	83.37	94.94	99.82	1.96
ACM	Office	87.60	97.90	100	1.71
	Mall	86.36	97.53	99.95	1.88
	Parking	81.72	95.44	99.56	2.12
	Mean	**85.23**	**96.96**	**99.84**	**1.90**

Table 2
The positioning results of shopping mall.

Phone Type	Path Length /m	CDF(3) /%		CDF(5) /%		CDF(10) /%		APE /m	
		BLE	MS-PF	BLE	MS-PF	BLE	MS-PF	BLE	MS-PF
Seq-01	780.51	63.04	77.62	85.63	97.33	98.36	100.00	2.96	2.22
Seq-02	448.85	62.11	93.19	90.37	99.69	99.38	100.00	2.69	1.67
Seq-03	718.21	75.22	87.43	96.41	96.77	100.00	100.00	2.25	1.90
Seq-04	534.57	70.07	89.07	93.59	98.57	99.52	100.00	2.59	1.85
Seq-05	741.83	67.45	71.5	93.09	93.09	99.83	100.00	2.51	2.41
Seq-06	747.84	65.09	80.73	91.15	93.05	100.00	99.53	2.57	2.40
Seq-07	548.50	84.83	93.79	97.93	99.31	100.00	100.00	1.91	1.37
Seq-08	573.36	62.05	89.55	93.18	100.00	100.00	100.00	2.71	1.66
Seq-09	527.08	72.59	94.32	96.05	100.00	100.00	100.00	2.40	1.45
Total/Mean	5620.75	69.16	**86.36**	93.04	**97.53**	99.68	**99.95**	2.51	**1.88**

4. Conclusion

This paper presents a handheld pedestrian indoor localization method utilizing a particle filtering framework that integrates IMU, magnetic sensors, BLE, and semantic maps. The method addresses unintentional hand movement interference by analyzing candidate steps detected in the time domain and applying a short-time Fourier transform in the frequency domain. Additionally, a magnetic interference coefficient is employed to model and quantify electromagnetic interference, thereby improving the accuracy of directional estimates across diverse indoor environments. The methodology further utilizes passable semantic information to model the indoor map, applying probabilistic constraints to various regions. This approach significantly enhances the robustness of the particle filtering system, yielding nearly a 10 % improvement in the Cumulative Distribution Function (CDF) at 3 m.

To assess the performance of the proposed method, extensive pedestrian positioning experiments were conducted using multiple handheld devices. In a shopping mall environment, the method achieved an average positioning accuracy of 1.88 m and a CDF(3) mean of 86.36 %, reflecting improvements of 17.20 % in accuracy and 25.10 % in robustness compared to Bluetooth-only methods. In a parking lot environment, the average positioning accuracy was 2.12 m, with a CDF(3) mean of 81.72 %, slightly lower than in the shopping mall. These results demonstrate that the proposed method provides high accuracy, long-term reliability, and robustness in large-scale and complex indoor environments.

Moreover, the method relies on pre-modeling and scene evaluation using laser SLAM equipment, which may not be feasible in all scenarios. Future research will aim to develop methods for modeling more reliable long-term semantic features in GPS-denied environments and to create more lightweight fusion algorithms to enhance performance, stability, and efficiency in navigation solutions.

CRediT authorship contribution statement

Xu Yang: Writing – review & editing, Writing – original draft, Validation, Methodology. **Xiaojuan Huang:** Validation, Formal analysis. **Yanshun Zhang:** Resources, Methodology, Funding acquisition. **Zhaoyang Liu:** Validation, Formal analysis. **Yang Pang:** Visualization, Software.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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